Reg No.:_______________ Name:__________________________

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

First Semester B.Tech Degree Examination December 2021 (2019 scheme)

Course Code: EST110

Course Name: ENGINEERING GRAPHICS

(**2019-Scheme**)

Max. Marks: 100 Duration: 3 Hours

Instructions: Retain Construction lines. Show necessary dimensions. Answer any ONE question from each module. Each question carries 20 marks

Module-I

1 The distance between the end projectors through the end points of line AB is 60 mm. The end A is 20 mm above HP and 15 mm in front of VP. The end B is 45 mm in front of VP and above HP. Front view of the line measures 75 mm. Draw the projections of line AB and find its true length and true inclinations with HP and VP.

2 The top view of a line PQ is 70 mm long and makes an angle of 45^0 with XY. The end P is in VP and 15 mm above HP. The end Q is 30 mm above HP and the whole line is located in first quadrant. Draw its projections and find its true length, length of its elevation, inclinations with reference planes and also locate its traces.

Module-II

3 A pentagonal pyramid of base edge 30 mm and axis length 60 mm is resting on VP on one of its base edges. The axis of the pyramid is inclined at 35^0 to VP and the resting base edge is inclined at 45^0 to HP. Draw the projection of the pyramid.

4 A right circular cone, 40 mm base diameter and 60 mm long axis is resting on HP on one point of base circle such that its axis makes 45^o inclination with HP and 40^o inclination with VP. Draw its projections.

Module-III

5 A cylinder with a 60 mm base diameter and 70 mm axis is resting on its base in the HP. It is cut by an auxiliary inclined plane which makes an angle of 60^0 with the HP and perpendicular to VP and passes through the top end of the axis. Draw its front view, sectional top view and true shape of the section.

6 A pentagonal prism of base 30 mm and axis 60 mm long is kept with its base on HP with a base edge perpendicular to VP. It is cut by a plane inclined at 45^0 to

HP, perpendicular to VP and passing through the mid point of the axis. Draw the development showing the remaining portion of the solid.

Module-IV

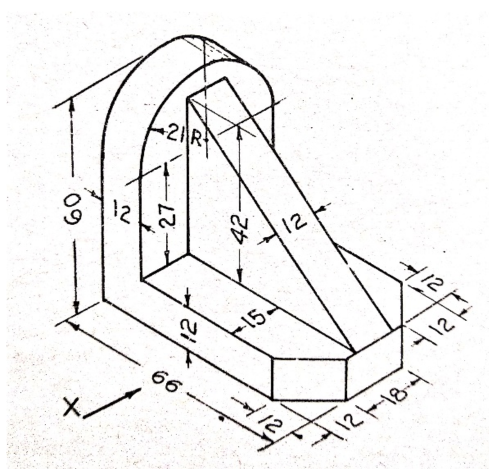
7 A sphere of 50 mm diameter is placed centrally on the top of the frustum of a square pyramid of 30 mm base side, 20 mm top side and the axis 50 mm long. Draw the isometric projection of the solids.

8 A hexagonal pyramid of base edge 25 mm and height 60 mm is surmounted centrally over a square slab of 70 mm side and 30mm thickness lying with its square side on HP so that one side of the square slab and one base edge of the pyramid are parallel to VP. Draw the isometric view of the combination.

Module-V

9 A square pyramid of base sides 30 mm and height 45 mm rests on its base on the ground with two base edges parallel to the Picture Plane (PP). The nearest edge of the base is 20 mm behind PP. The station point is situated at a distance of 70 mm in front of the PP, 40 mm to the right of the axis of the pyramid, and 60 mm above the ground. Draw the perspective view of the pyramid.

10 Draw the top view, front view and any one side view of the figure shown below. The front view direction is marked as X. Any missing dimension may be suitably assumed.



20X5=100 marks

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY First Semester B.Tech Degree Examination December 2021 (2019 scheme)

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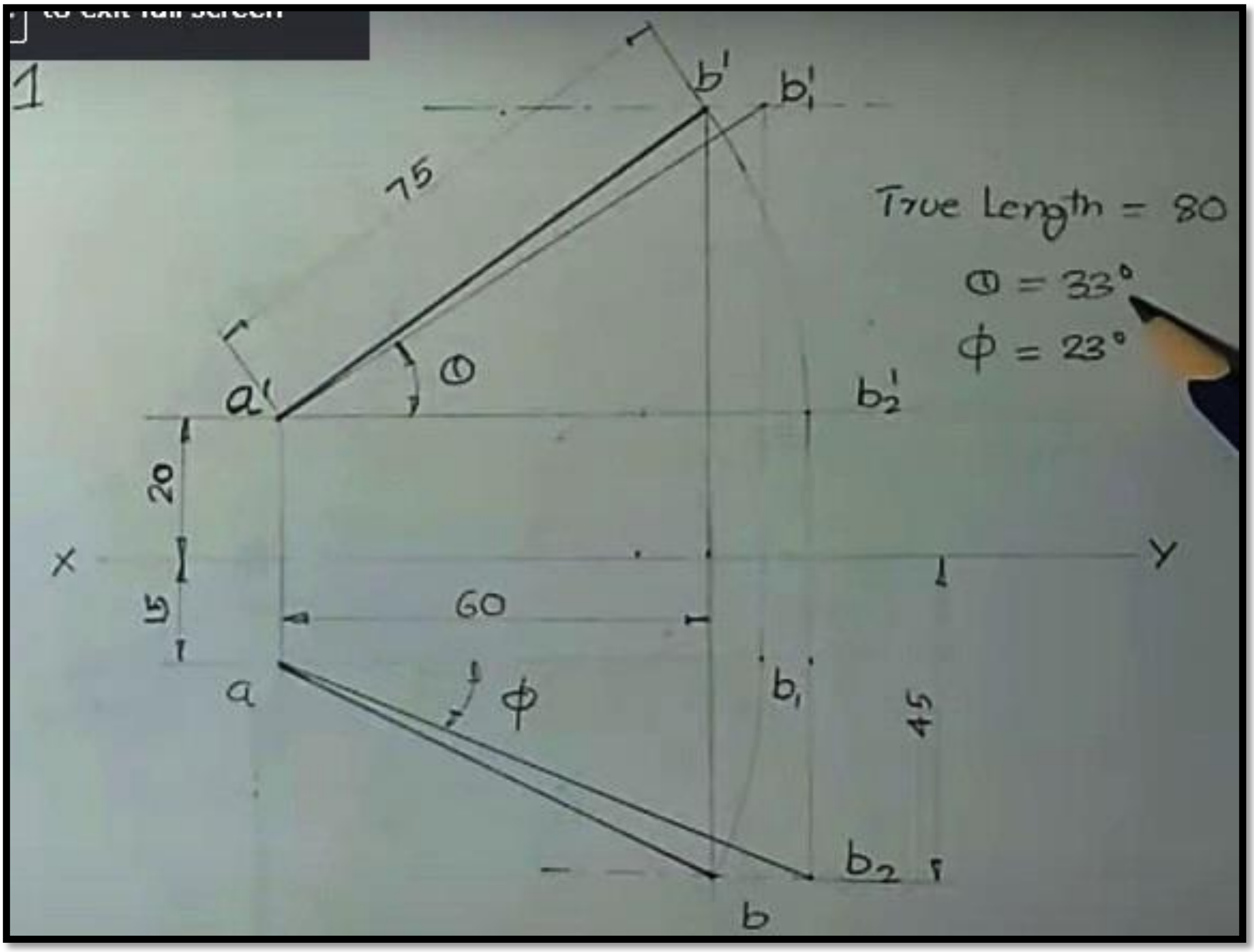
Answer Key

1.



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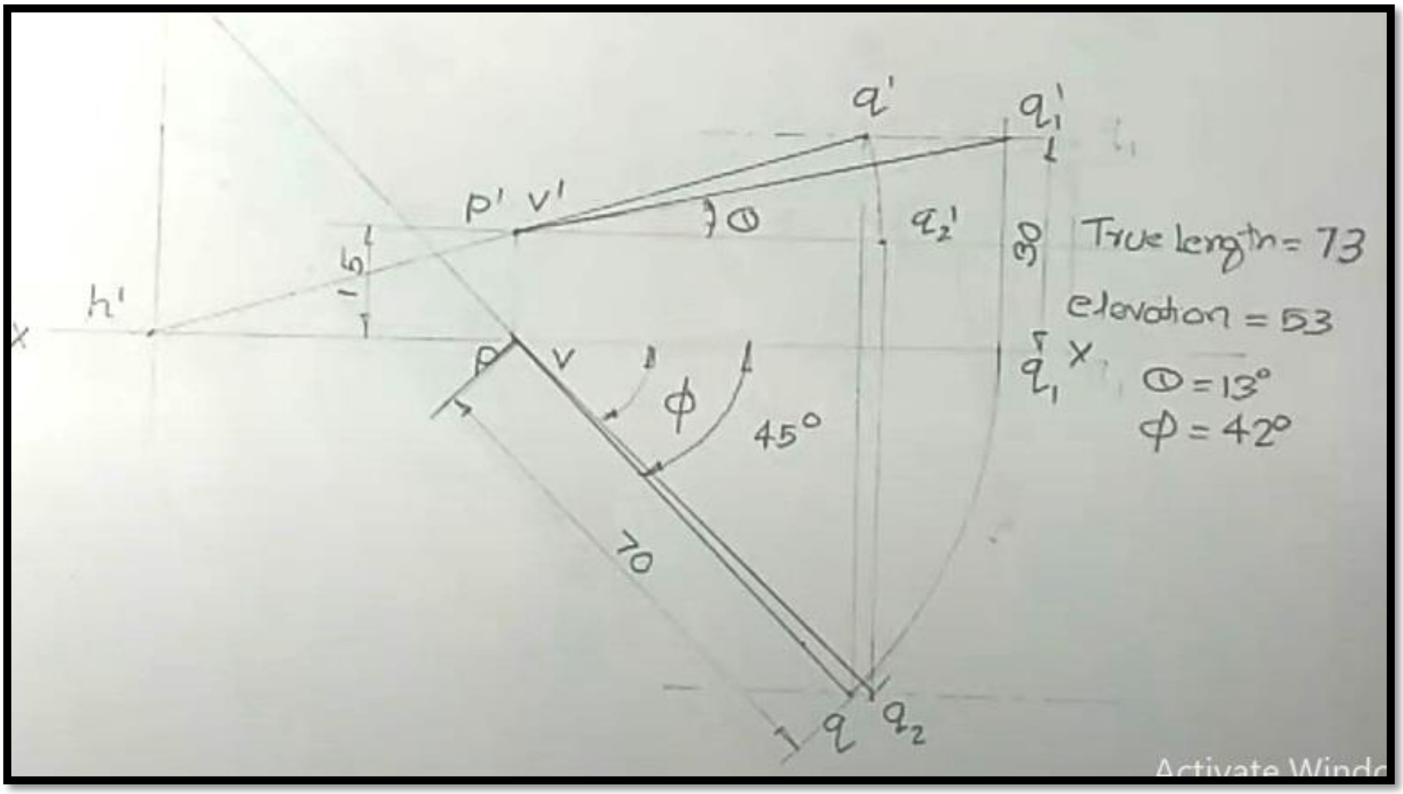
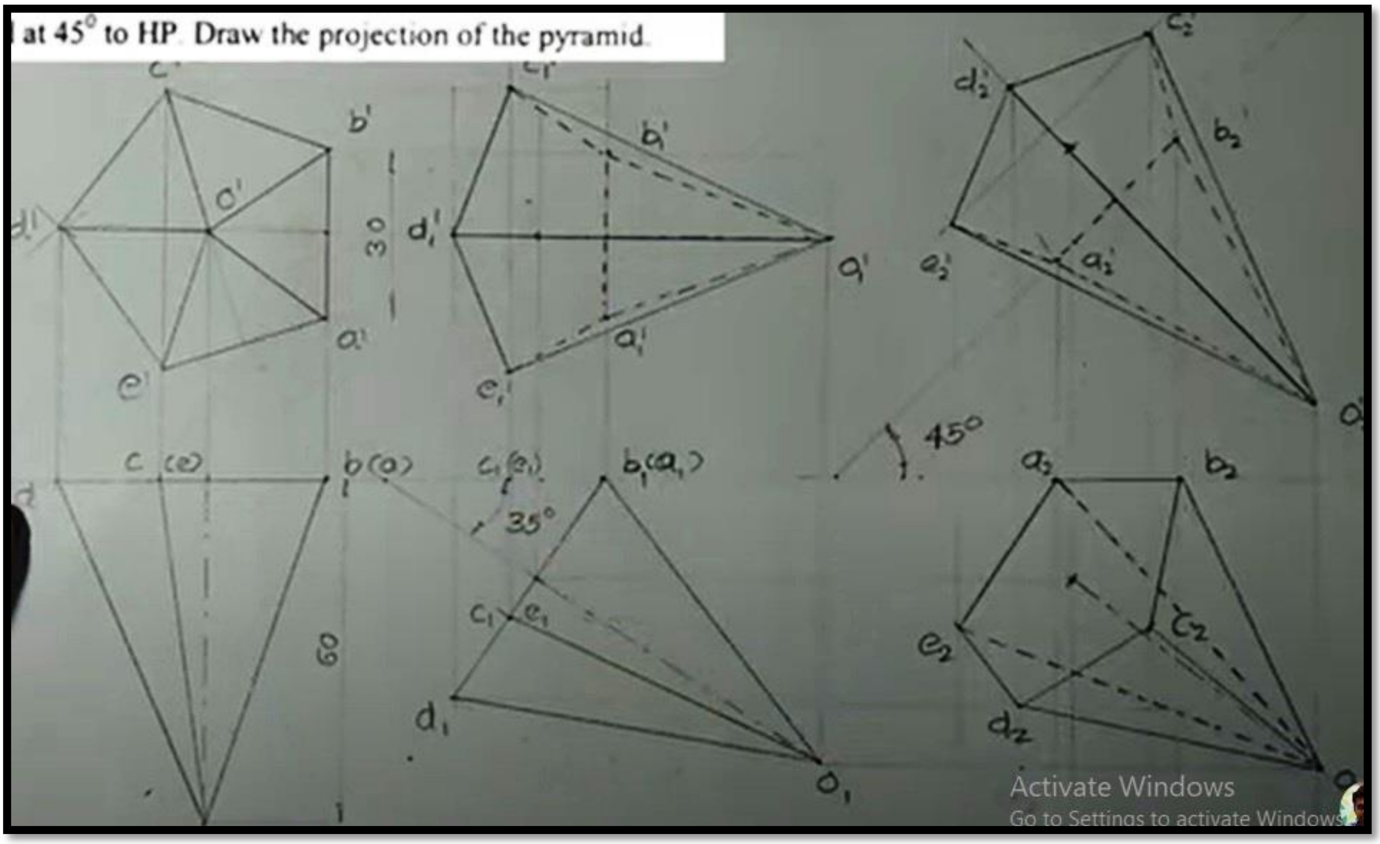


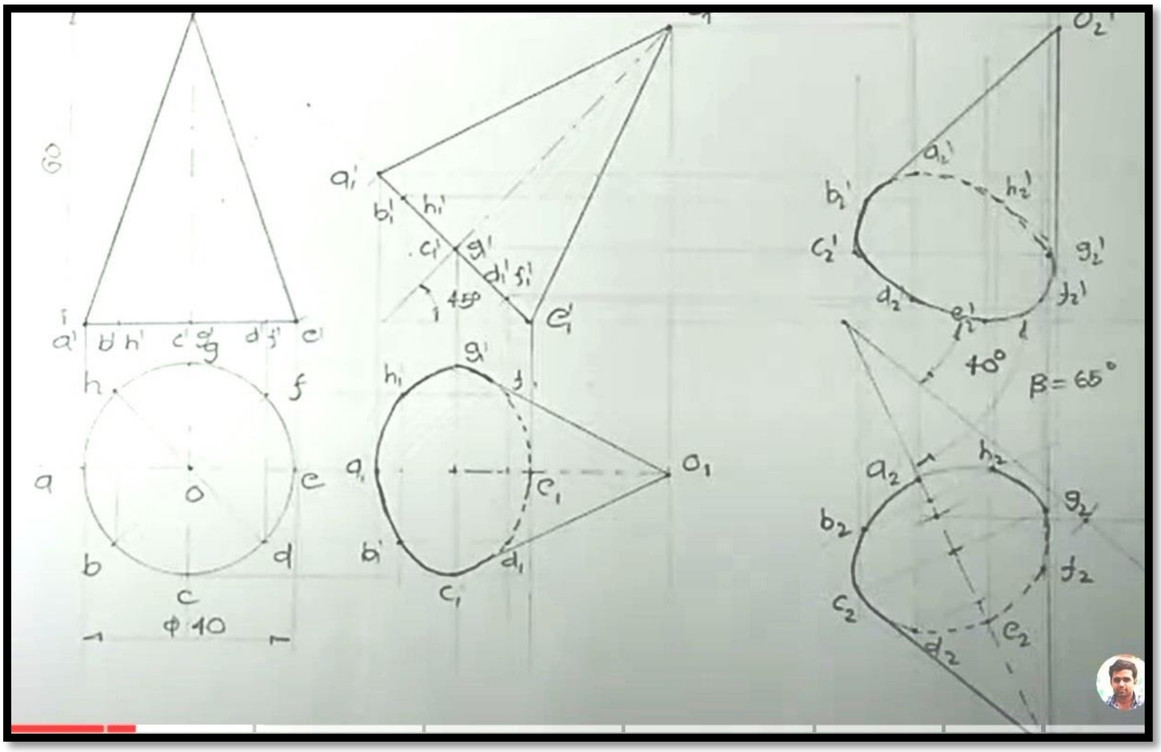
q'
q_1'
p' v'
θ
q_2'
30
True length = 73
elevation = 53
h'
15
x
p
v
φ
45°
q_1
y
θ = 13°
φ = 42°
70
q
q_2

3.



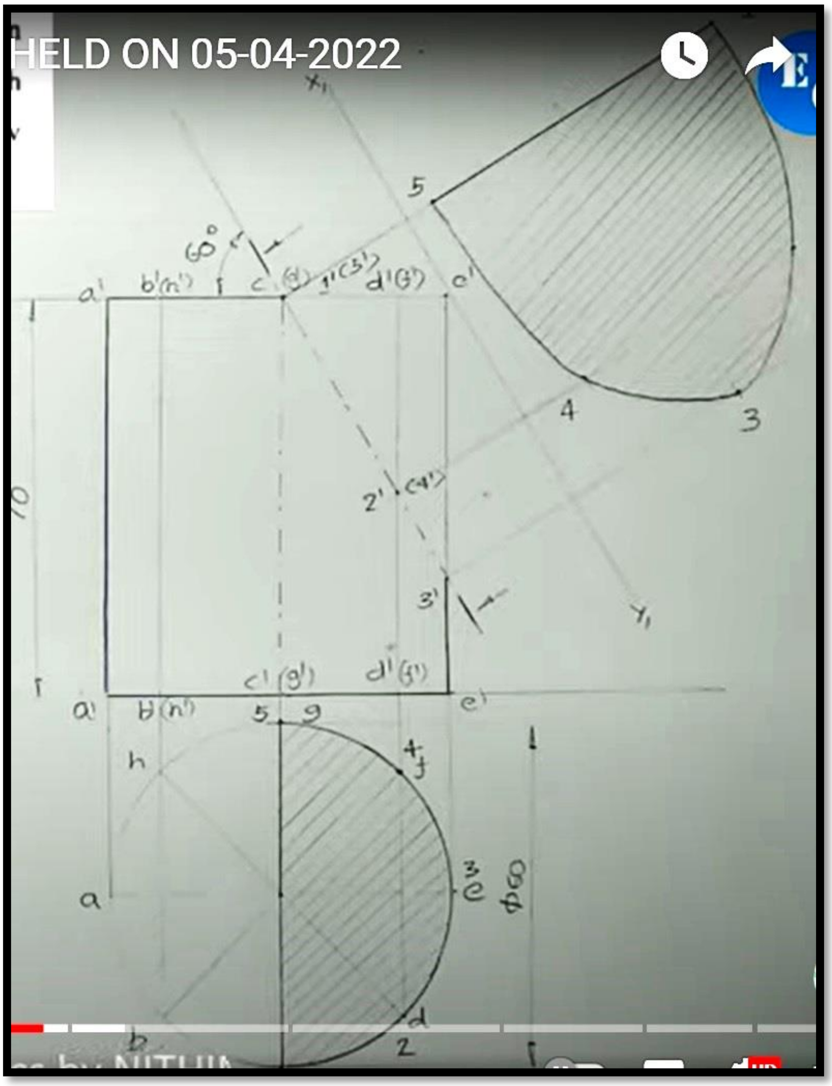
at 45° to HP. Draw the projection of the pyramid.
c'
b'
o'
d'
30
d_1'
b_1'
c_1'
o'
a'
e'
a_1'
e_1'
d_2'
c_2'
b_2'
e_2'
a_2'
45°
c (e)
b (a)
$c_1(e_1)$
$b_1(a_1)$
35°
60
c_1
e_1
d_1
o_1
a_2
b_2
c_2
e_2
d_2
Activate Windows
Go to Settings to activate Windows

4.



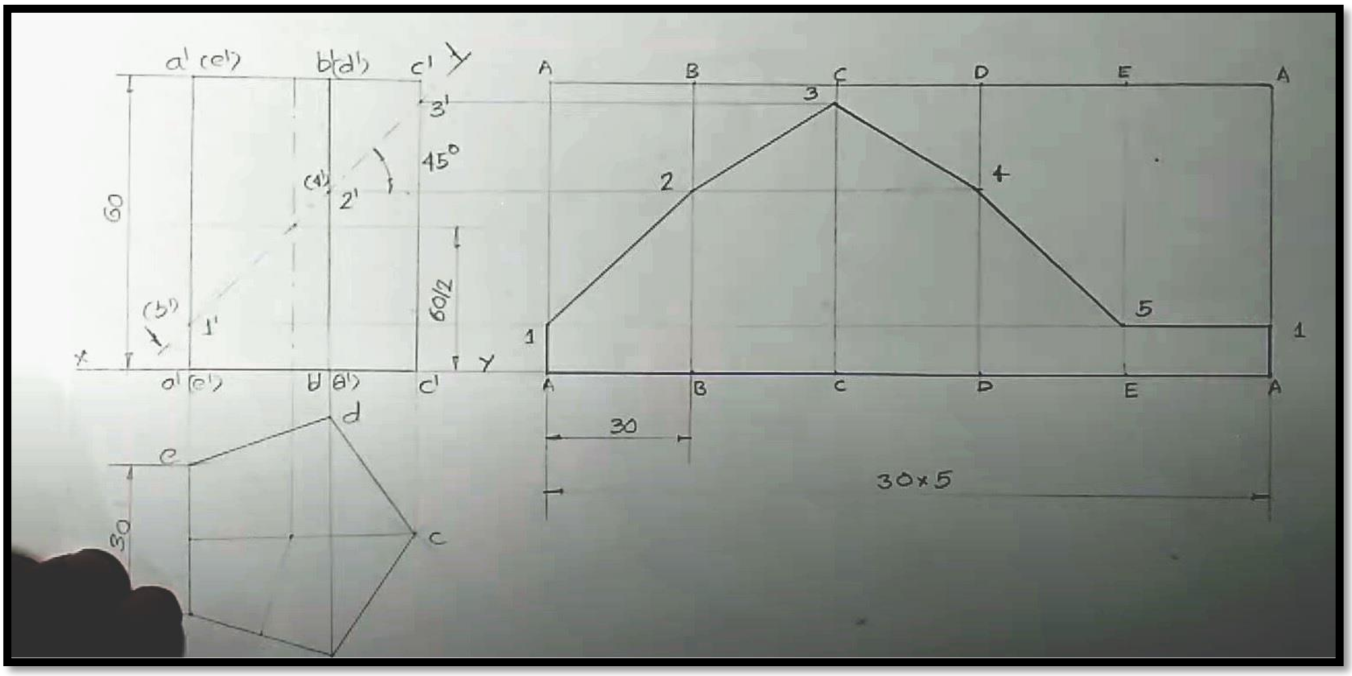
60
a' b' h' c' g' d' f' e'
h
f
a
o
e
b
d
c
ϕ 40
a1'
b1' h1'
c1' g1'
d1' f1'
45°
e1'
g1'
h1
f1
a1
e1
o1
b1
d1
c1
o2'
a2'
h2'
b2'
c2'
g2'
d2'
e2'
f2'
40°
β = 65°
a2
h2
b2
g2
f2
c2
e2
d2

5.

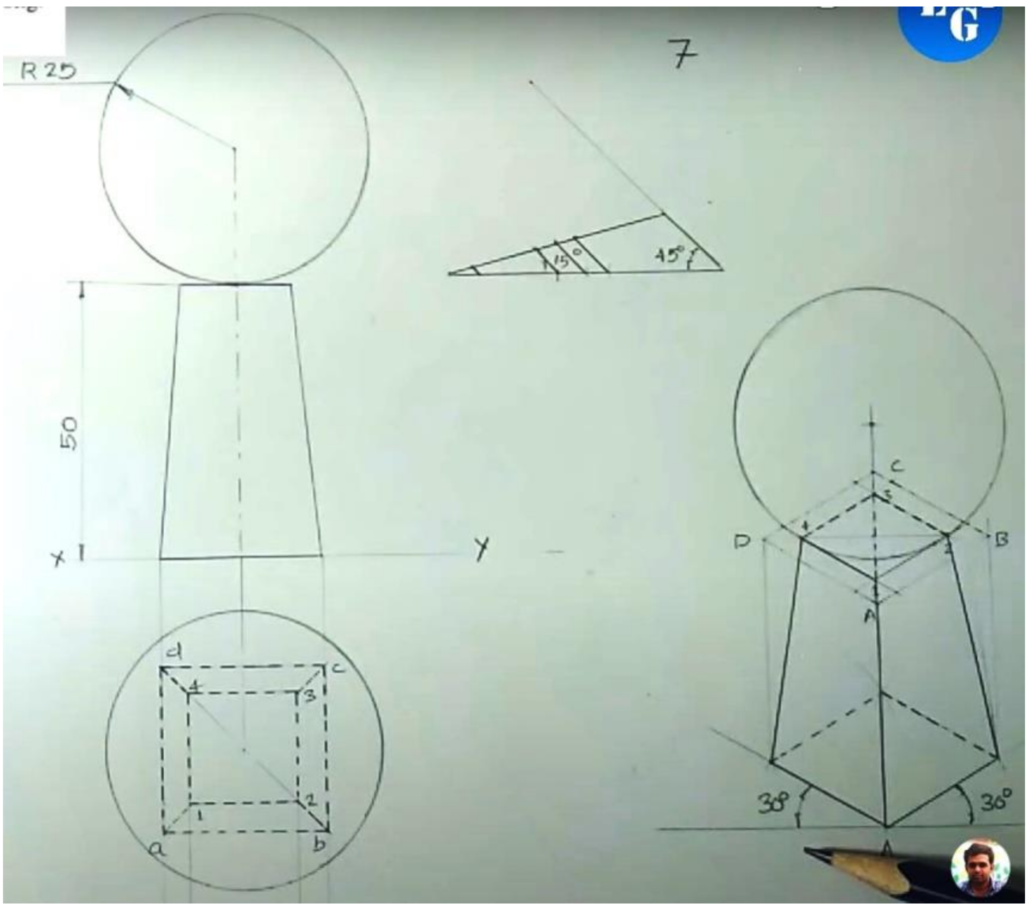


HELD ON 05-04-2022
x1
5
60°
a' b'(h') c' d'(g') e'
4
3
2' (4')
3'
y1
c'(g') d'(f')
a' b'(h') 5 g e'
h
4 f
a
3 e
ϕ 60
d
2

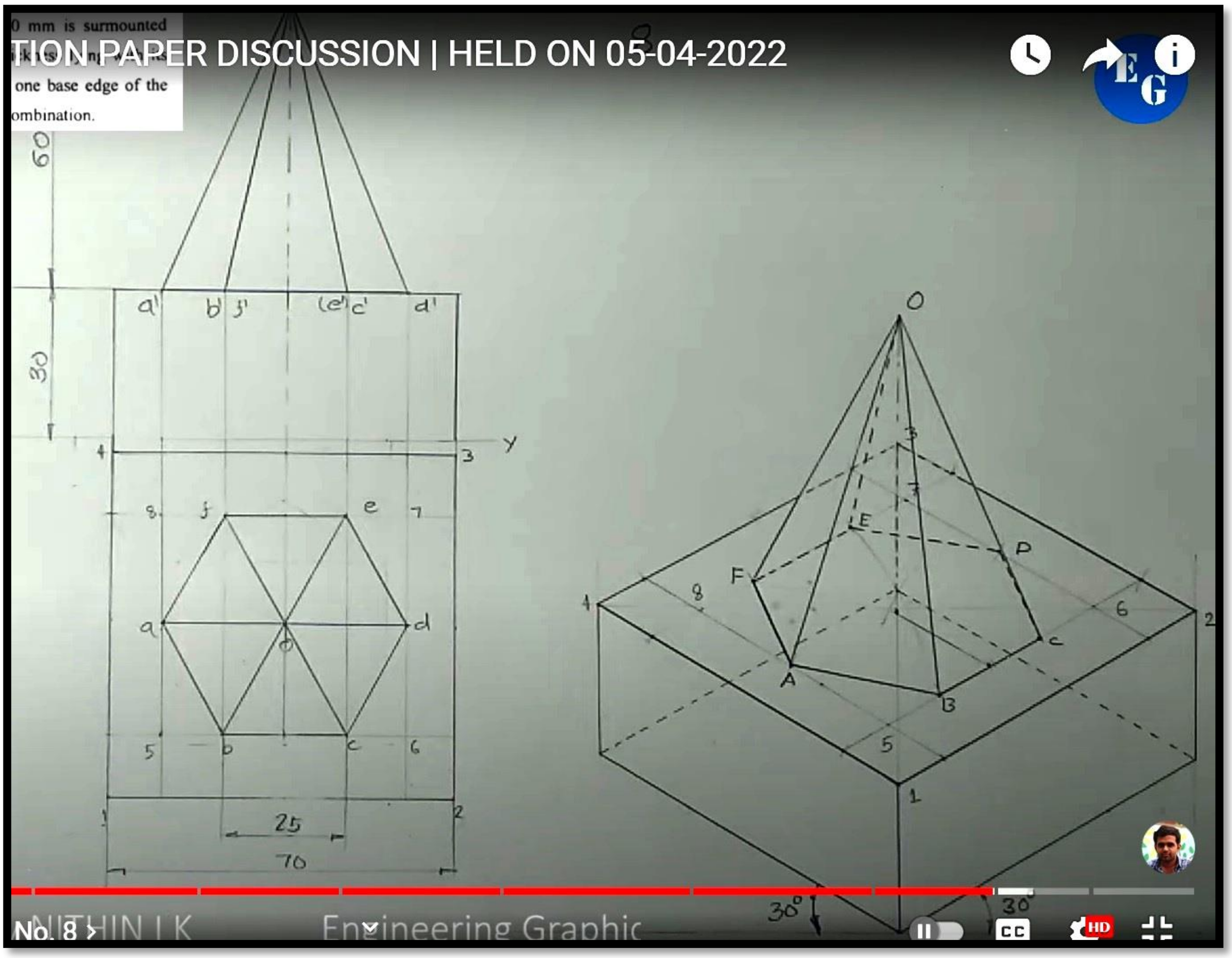
6.



7.

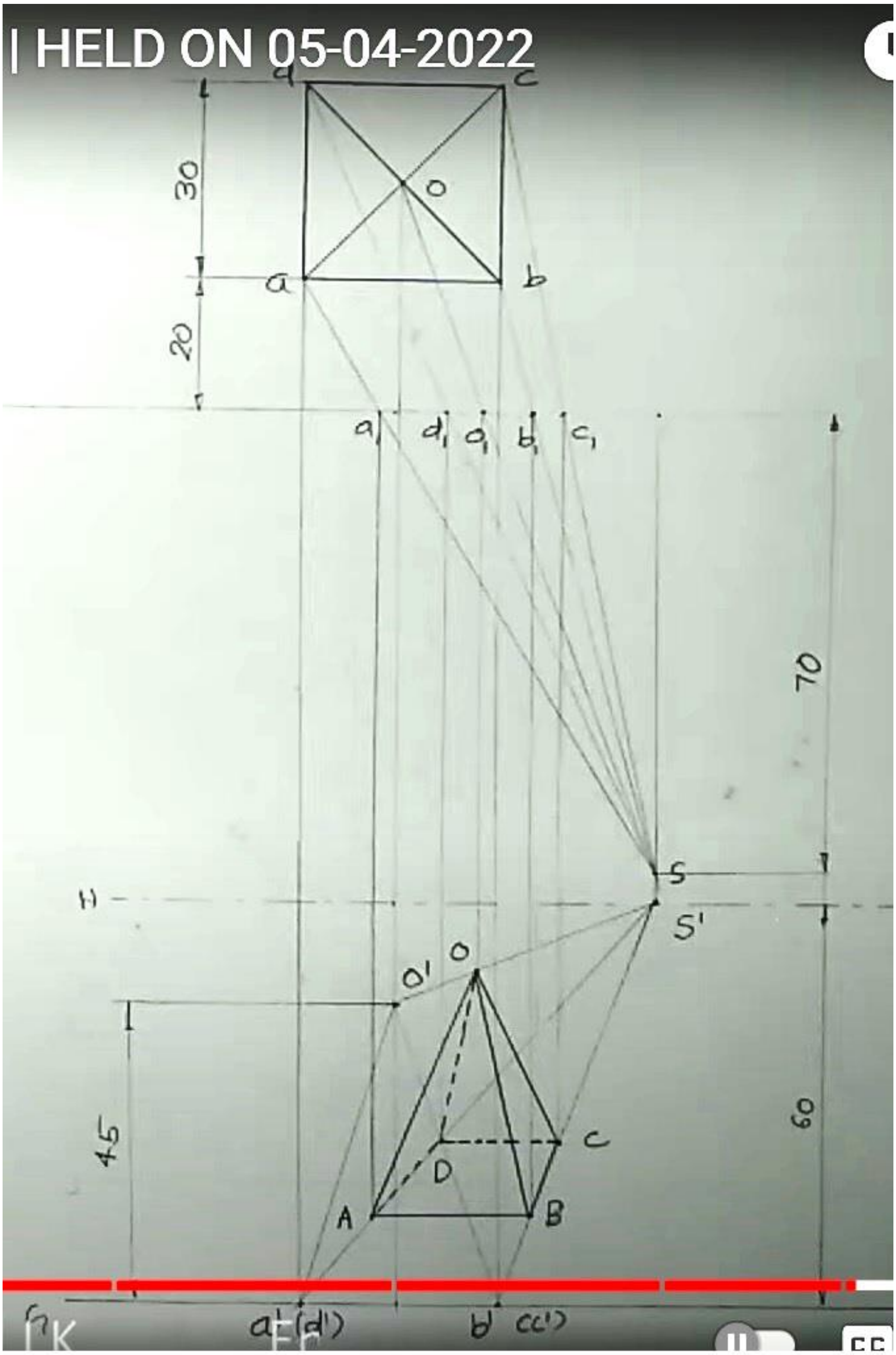


8.

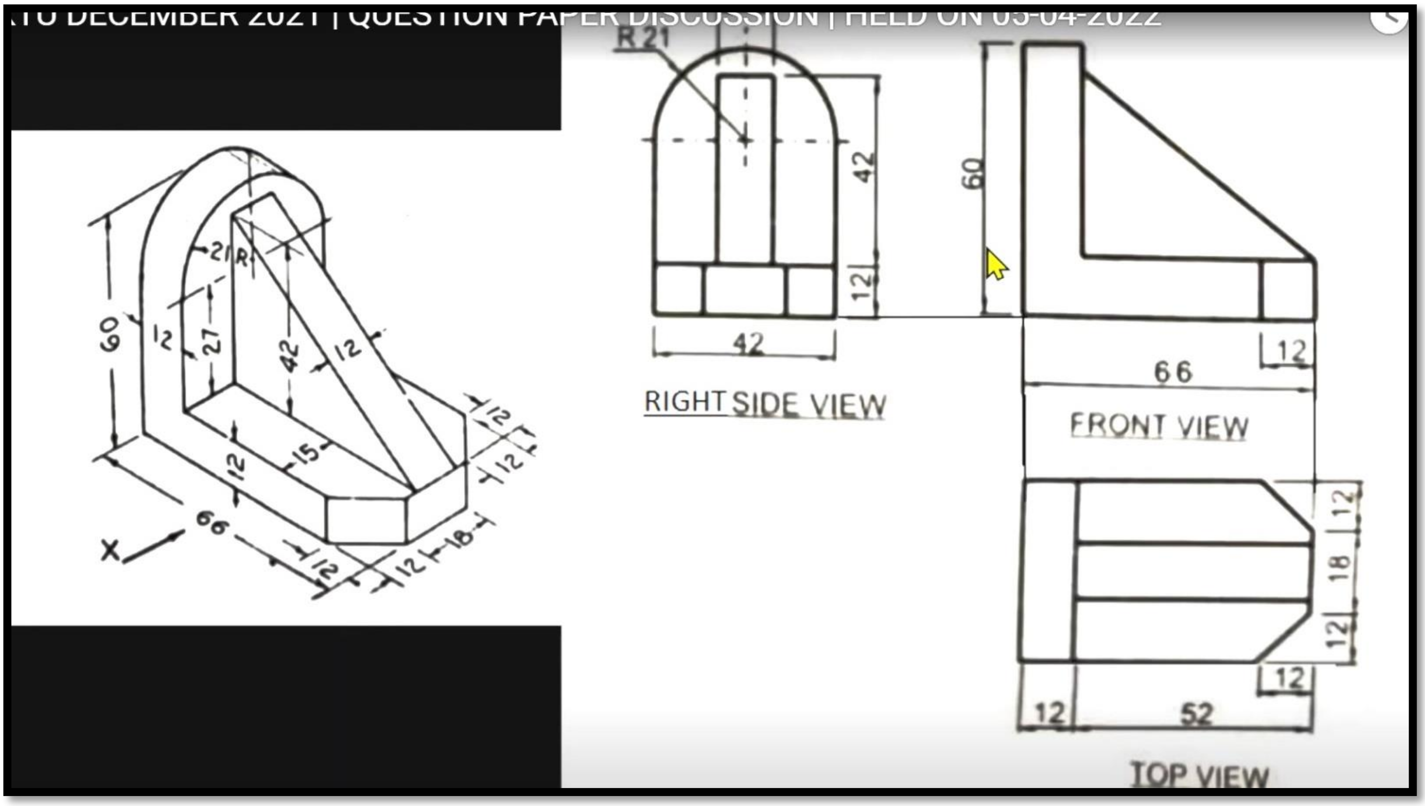


0 mm is surmounted
one base edge of the
ombination.
TION PAPER DISCUSSION | HELD ON 05-04-2022
60
30
a'
b' f'
(e')c'
d'
y
4
3
8
f
e
7
a
d
o
5
b
c
6
1
2
25
70
O
E
F
P
A
B
C
8
6
5
1
2
30°
30°
No. 8
NITHIN I K
Engineering Graphic

9.


| HELD ON 05-04-2022
d
c
30
o
a
b
20
a1
d1
o1
b1
c1
70
S
H
S'
O
O'
45
60
C
D
A
B
a'(d')
b' (c')

10.



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Scheme/ Answer Key for Valuation

Scheme of evaluation (marks in brackets) and answers of problems/key

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

FIRST SEMESTER B.TECH DEGREE EXAMINATION, DECEMBER 2021

Course Code: EST110

Course Name: ENGINEERING GRAPHICS

(**2019-Scheme**)

Max. Marks: 100 Duration: 3 Hours

Instructions: Retain necessary Construction lines. Show necessary dimensions. Answer any ONE question from each module. Each question carries 20 marks

Module-I

1 Locating the front views and top views of end point: 6 Marks

Front View of the Line: 2 Marks

Top View of the Line: 5 Marks

True Length and Inclinations with HP and VP: 5 Marks

Dimensioning and Neatness: 2

2 Locating the end point p’ and p– 4 marks

Drawing the locus of q’– 2 marks

Drawing the top view – 2 marks

Drawing the front view – 2 marks

Drawing the projection by any one method – 2 marks

Finding true length and elevation length of the line – 2 marks

Finding true inclination with HP & VP – 2 marks

Locating HT & VT – 2 marks

Dimensioning and neatness – 2 marks

Module-II

3 Drawing initial position plan and elevation – 4 marks

First inclination views – 4 marks

Second inclination views -8 marks

Marking invisible edges – 2 marks

Dimensioning and neatness – 2 marks

4 Simple position - 5 marks, first rotation – 6 marks, second rotation- 7 marks, Dimensions and neatness- 2 marks

Module-III

5 Simple position front view and top view-5, section plane-3, sectional top view-5, true shape-5, neatness and dimension-2

6 Drawing initial position plan and elevation– 5 marks

Locating section plane as per given condition-4 marks

Development of the prism-5 marks

Development of the bottom portion- 5 marks

Dimensioning-1mark

Module-IV

7 **If answer is correct and simple position is not drawn distribute the marks equally between the isometric projection of pyramid and sphere**

Drawing projection in simple position-4 Marks

Isometric scale-3 Marks

Isometric projection of square pyramid-6 Marks

Isometric projection of sphere-5 Marks

Dimensioning, marking of points and neatness-2 Marks

8 **If answer is correct and simple position is not drawn distribute the marks equally between the isometric view of pyramid and slab**

Drawing initial position plan and elevation– 4 marks

Isometric View of pyramid – 7 marks

Isometric View of slab – 7 marks

Dimensioning and neatness – 2 marks

Module-V

9 Front View: 3Marks

Top View: 3 Marks

Location of Station Points: 2 Marks

Perspective View:10 Marks

Dimensioning and Neatness: 2 Marks

10 Front View: 6 Marks

Top View: 6 Marks

Profile View: 5 Marks

Dimensioning and Neatness: 3

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